

Business Contract Validation System

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1. Introduction

1.1 Purpose

The Business Contract Validation System automates and streamlines the verification of business contracts using AI-powered tools. The system extracts text from scanned contract images, preprocesses it, and applies a Convolutional Neural Network (CNN) model for validation. It then utilizes APIs such as Google Gemini for further legal clause analysis and verification, ensuring compliance with legal standards and regulations.

1.2 Document Conventions

This document follows the IEEE SRS 830-1998 format, ensuring a structured and standardized specification.

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- **Software Developers:** Implementing the system and integrating AI and API functionalities.
- **Business Analysts:** Understanding the business needs and use cases.
- **AI Engineers:** Enhancing OCR, NLP, and CNN model performance.
- **Legal Professionals:** Validating the correctness and compliance of contract data.
- **Project Managers:** Overseeing system development and implementation.

1.4 Scope

The Business Contract Validation System ensures contracts meet legal and compliance requirements by performing the following steps:

- Image preprocessing: Grayscale conversion to improve OCR accuracy.
- OCR-based text extraction: Extract text from scanned contract images.
- Text preprocessing: Using NLP (SpaCy) for lemmatization and stopword removal.
- Contract validation using CNN: Classifying contracts as valid/invalid based on learned patterns.
- API-based legal clause verification: Cross-checking contract clauses with external AI services.
- User interface for review and validation: Providing a dashboard for contract submission, validation results, and insights.

1.5 References

- IEEE 830-1998 SRS Guidelines
 - OpenCV, Tesseract OCR, and SpaCy Documentation
 - TensorFlow/Keras for CNN Model
 - Google Gemini API Documentation
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2. Overall Description

2.1 Product Perspective

The Business Contract Validation System integrates AI-based document processing, NLP-based text analysis, and deep learning models to automate contract validation. It aims to replace manual verification processes and reduce legal risks by providing accurate compliance assessments.

2.2 Product Functions

- **Image Processing:** Convert contract images to grayscale for enhanced OCR performance.
- **OCR Text Extraction:** Extract contract text using Tesseract OCR.
- **Text Preprocessing:** Tokenization, lemmatization, and stopword removal using SpaCy.
- **CNN-Based Contract Validation:** Classify contracts as Valid or Invalid based on predefined rules.
- **API-Based Clause Verification:** Utilize external APIs (e.g., Google Gemini) to analyze legal clauses.
- **User Dashboard:** Interface for contract submission, validation results, and recommendations.

2.3 User Characteristics

- **Legal Analysts:** Require clause validation and risk assessment.

- Business Managers: Need contract compliance reports and decision support.
- Developers: Maintain system performance and AI model efficiency.
- End-Users: Interact with the system through an intuitive UI.

2.4 Constraints

- Requires high-resolution scanned contract images for optimal OCR performance.
- Accuracy depends on training data and CNN model robustness.
- External API response times may introduce processing delays.
- Compliance regulations may vary across jurisdictions, requiring API updates.

2.5 Assumptions and Dependencies

- Uses Google Gemini API or similar for legal analysis.
 - Requires a GPU for CNN training to improve classification efficiency.
 - Relies on SpaCy and TensorFlow for text processing and deep learning.
-

3. Specific Requirements

3.1 Functional Requirements

3.1.1 Image Preprocessing

- Convert contract images to grayscale to enhance OCR accuracy.
- Objective: Enhance the quality of contract images for OCR (text extraction).
- Purpose: OCR software, like Tesseract, performs better on grayscale images, as it eliminates unnecessary color information and focuses solely on text, improving text detection accuracy.
- Tech: Python libraries like OpenCV can be used to convert color images to grayscale.
- Resize and normalize images:
- Purpose: Resizing ensures that all images are of a uniform size, which helps OCR models perform better. Normalizing adjusts the image's pixel values (usually to a range of 0-255) to enhance the consistency and performance of OCR algorithms.
- Tech: OpenCV or Pillow can be used for resizing, and NumPy can be employed for pixel normalization.

3.1.2 OCR Text Extraction

- Extract text from images using Tesseract OCR.
 - Store extracted text in a structured format for analysis.
- Tesseract OCR:
- Purpose: Tesseract is an open-source Optical Character Recognition engine that can recognize text from images and convert it into machine-readable format.
 - Tech: Tesseract is often used in Python via the pytesseract library.

- The extracted text will be stored in a structured format (like JSON or CSV) for easier analysis later on.

3.1.3 Text Preprocessing

- Tokenization, lemmatization, and stopword removal using SpaCy NLP.
- Objective: Clean and prepare the extracted text for further analysis.

Tokenization:

- Purpose: Break the text into smaller chunks, such as words or phrases, for further processing. This is important for analyzing the meaning and structure of the contract text.
- Tech: SpaCy is a popular library in Python that provides easy-to-use tokenization functions.

Lemmatization:

- Purpose: Convert words into their base form (e.g., "running" becomes "run"). This helps with consistent analysis and reduces complexity in the contract's textual data.
- Tech: SpaCy provides lemmatization as part of its NLP toolkit.

Stopword Removal:

- Purpose: Remove common but irrelevant words (like "the", "and", "in") that don't add much meaning to the analysis.
- Tech: SpaCy comes with a pre-defined list of stopwords that can be filtered out during preprocessing.

Noise Removal:

- Purpose: Clean the extracted text by removing unwanted characters, such as symbols, excessive punctuation, or formatting errors.
- Tech: Regular expressions (using re in Python) can be used to detect and remove unwanted characters.

3.1.4 CNN-Based Contract Validation

- Train a CNN model to classify contracts as Valid/Invalid and detect missing clauses and highlight inconsistencies.
- Objective: Use a Convolutional Neural Network (CNN) to classify contracts as valid or invalid.

Train a CNN model:

- Purpose: The CNN is trained to learn patterns from labeled data (valid vs invalid contracts) to classify new contracts based on whether they are legally correct.
- Tech: TensorFlow or PyTorch can be used to create and train the CNN model. These libraries help build the deep learning model, train it on the dataset, and validate its performance.

Detect Missing Clauses and Inconsistencies:

- Purpose: The CNN model can be extended to identify specific issues in contracts, such as missing clauses or inconsistencies between sections.
- Tech: Techniques like transfer learning (using pre-trained models like ResNet or Inception) can be used to fine-tune the CNN for contract-specific validation tasks.

3.1.5 API-Based Validation

- Send extracted text to AI APIs (Google Gemini) for legal verification.

Google Gemini (or other AI APIs):

- Purpose: Google Gemini or similar legal AI APIs analyze the text for compliance with legal standards, risk assessments, and potential issues in the contract.

- Tech: APIs like Google Cloud Natural Language API or specialized legal verification APIs can be used to send the extracted text for analysis. The API will return data such as:
- Compliance status: Whether the contract adheres to required legal standards.
- Risk assessment: Evaluation of the contract's risks.
- Suggested modifications: Proposals to improve or correct the contract.

3.1.6 User Interface

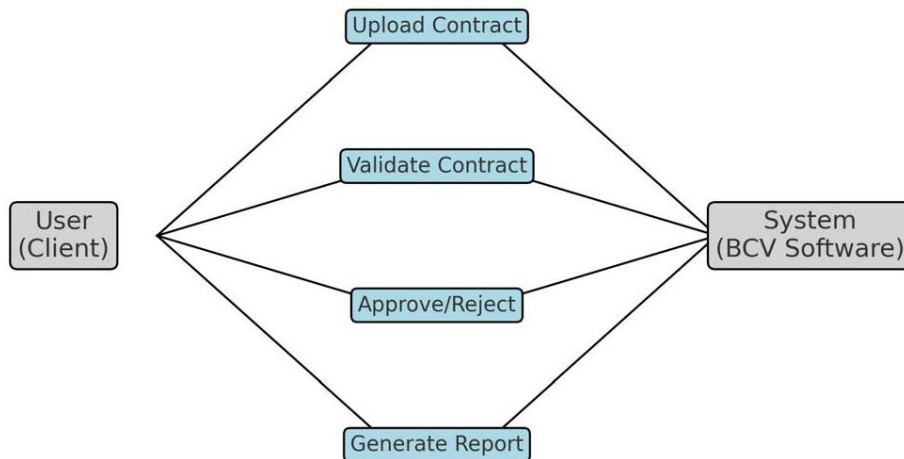
- Allow users to upload contract documents.
- Display validation reports, detected risks, and recommendations.
- Provide downloadable results for further legal assessment.

3.2 Non-Functional Requirements

- Scalability: Must handle multiple contract validations simultaneously.
 - Accuracy: CNN model should achieve >90% accuracy in contract classification.
 - Security: Contract data must be encrypted and comply with GDPR.
 - Performance: OCR processing should complete in <3 seconds per contract.
 - Availability: System uptime should be 99.9% to ensure continuous validation.
-

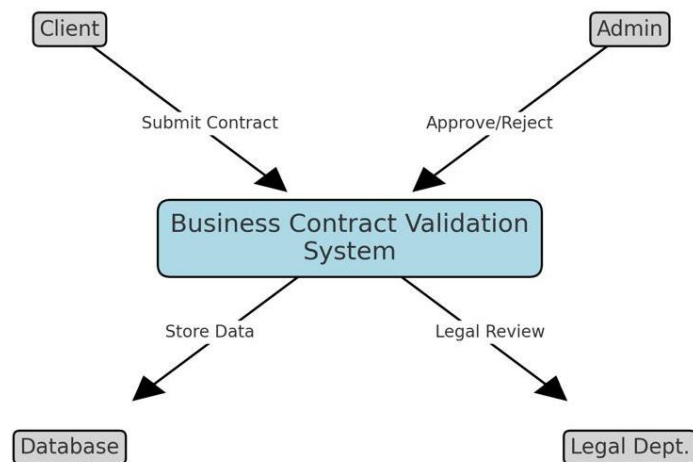
4. Project Diagram

4.1 Use-Case Diagram

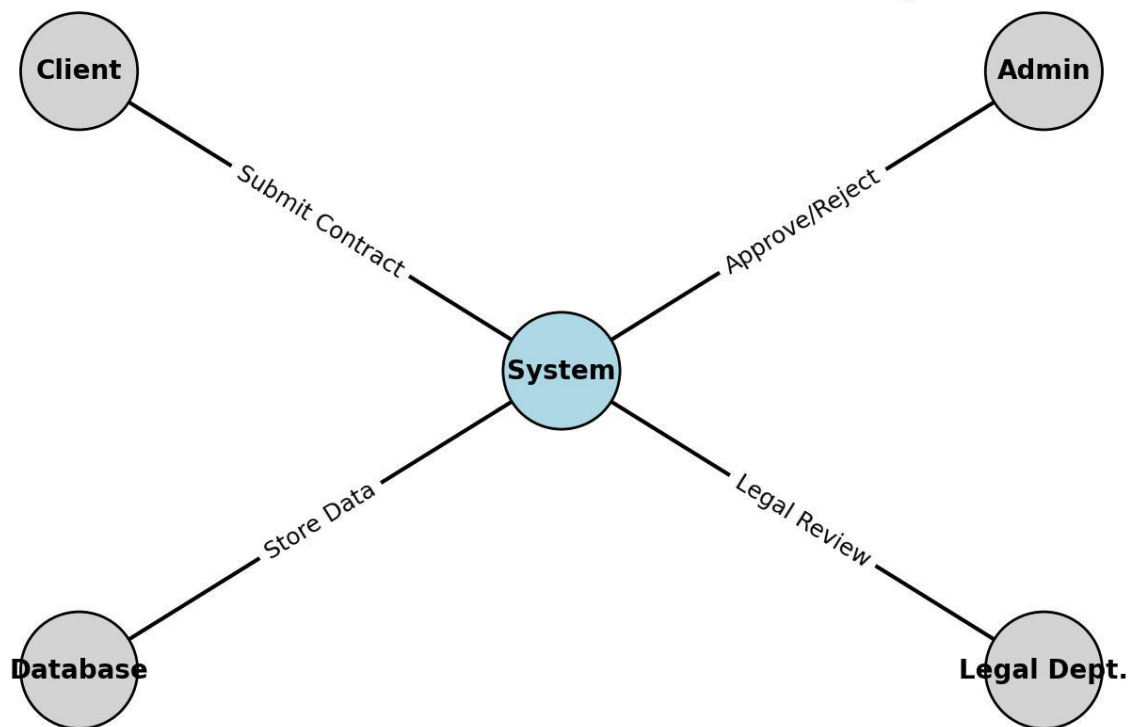


4.2 DFD_diagram:

Level 0 DFD- BUSINESS CONTRACT VALIDATION SYSTEM

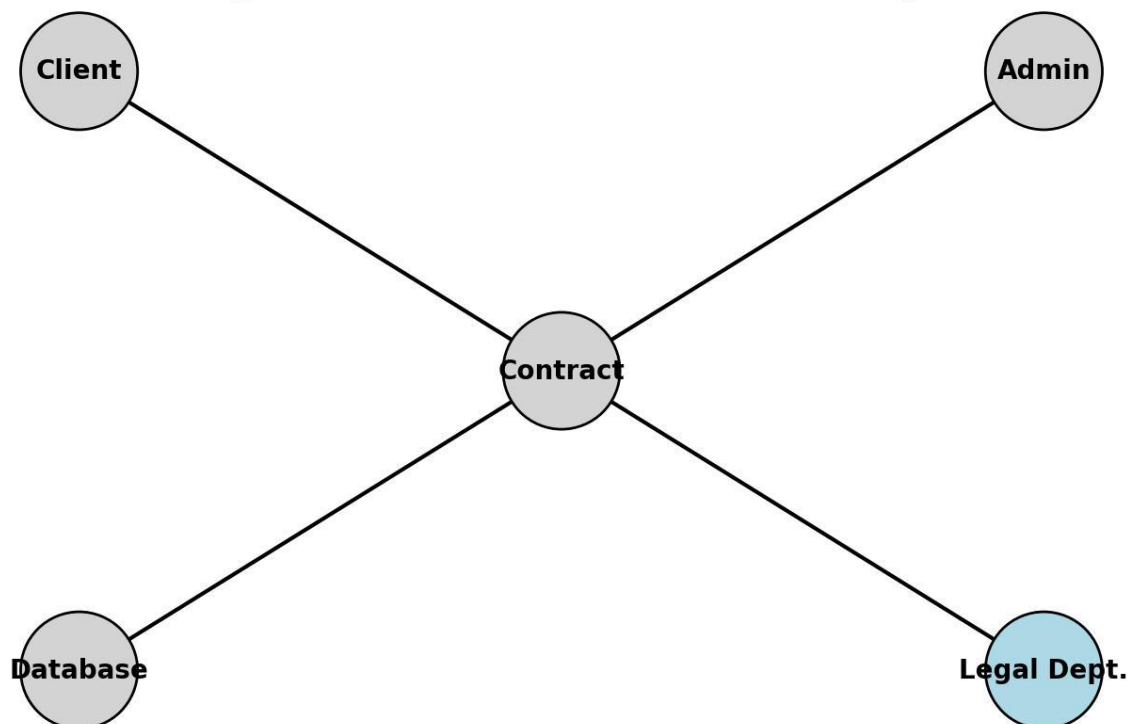


Level 1 DFD - Business Contract Validation System



4.3 ER_diagram:

ER Diagram - Business Contract Validation System



5. Appendices

5.1 Acronyms and Abbreviations

- OCR: Optical Character Recognition
- CNN: Convolutional Neural Network
- API: Application Programming Interface
- NLP: Natural Language Processing
- IEEE: Institute of Electrical and Electronics Engineers

5.2 References

- Google Gemini API Documentation
- IEEE SRS 830-1998 Standard
- TensorFlow/Keras Documentation
- OpenCV & Tesseract OCR Guides